BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 059 · Lesson 059 · Thinking Abstractly: What's the Model?

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What makes a good field in a record?

Q2. Why give a unique record key?

Q3. What is a precondition?

Q4. 025

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

abstraction — Focus on relevant details.

model — Simplified representation of reality.

assumption — Condition accepted within a model.

irrelevant — Detail that does not affect task.

New learning

Understand — Abstraction selects only the information required for the task and deliberately excludes irrelevant real-world detail.

Model — An abstract model is a simplification of reality: here distance and child/adult status determine the fare; hair colour does not.

Evaluate — The model is only as good as its assumptions. If peak pricing or concessions apply, the abstraction needs refinement.

Worked example

Choose relevant inputs for calculating parking fee at £2 per hour, rounded to whole hours.

1. Explain — ParkingFee(hours)=hours×£2 for the simplified model; define non-negative hours and rounding rules.

We Do · Practise with guidance

Model a library overdue fee of 20p per late day; decide which real-world details matter.

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Input lateDays (non-negative); rate £0.20/day; output fee. Book cover colour
  is irrelevant.
```

Explain

Fee=0.20×lateDays in this simplified rule; real libraries might cap fees or exempt users.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

For an automated bus fare model, identify three relevant and two irrelevant details. [5]

Question 2

Define inputs, processing and output for library overdue-fee model at 20p/day capped at £2. [6]

Question 3

Evaluate the risks of abstracting a school attendance intervention down to attendance percentage alone. [6]

Exit Ticket · Answer before checking

An abstraction must include every observable detail of reality.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Information needed for the model/use case.

Q2. To identify instances unambiguously.

Q3. Assumption that must hold for a calculation to be valid.

Q4. Searching target6 in sorted [2,4,6,8,10]: which mid index is checked first, and result?

You Do pages · Independent questions

Q1. Relevant child category, distance, fare rules [3]; clothing colour, phone brand etc irrelevant [2].

Q2. Input daysLate [1] non-negative validation [1]; fee=MIN(0.20×days,2.00) [3]; output fee [1].

Q3. Simple measurable indicator [1] but misses illness/SEND/context [2]; can bias decisions [1]; human review and further relevant variables [2].

Exit Ticket · Model answer

False. It deliberately excludes irrelevant detail.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first.

GREEN PEN Then: tick and rewrite mistakes.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding